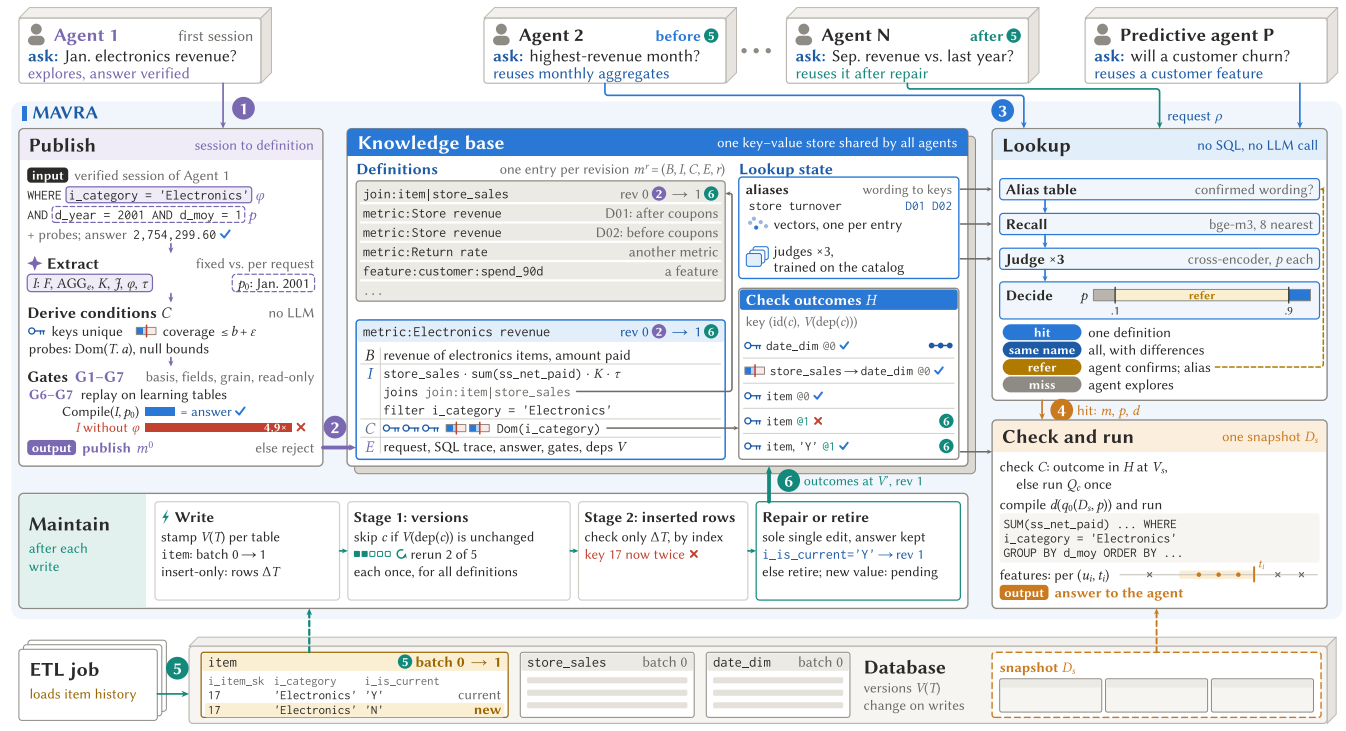

Agent 1
first session
ask: Jan. electronics revenue?
explores, answer verified
Agent 2
before 5
ask: highest-revenue month?
reuses monthly aggregates
Agent N
after 5
ask: Sep. revenue vs. last year?
reuses it after repair
Predictive agent P
ask: will a customer churn?
reuses a customer feature
MAVRA
1
Publish
session to definition
input
verified session of Agent 1
WHERE i_category = 'Electronics'
AND d_year = 2001 AND d_moy = 1
+ probes; answer 2,754,299.60
Extract
fixed vs. per request
I: F, AGG_e, K, J, φ, τ
ρ0: Jan. 2001
Derive conditions C
no LLM
keys unique
coverage ≥ b + ε
probes: Dom(T. a), null bounds
Gates G1–G7
basis, fields, grain, read-only
G6–G7 replay on learning tables
Compile(I, p0)
= answer
I without φ
4.9×
output
publish m0
else reject
2
Knowledge base
one key–value store shared by all agents
Definitions
one entry per revision m^r = (B, I, C, E, r)
join:item|store_sales
rev 0 2 → 1 6
metric:Store revenue
D01: after coupons
metric:Store revenue
D02: before coupons
metric:Return rate
another metric
feature:customer:spend_90d
a feature
...
metric:Electronics revenue
rev 0 2 → 1 6
B
revenue of electronics items, amount paid
I
store_sales · sum(ss_net_paid) · K · τ
joins join:item|store_sales
filter i_category = 'Electronics'
C
Dom(i_category)
E
request, SQL trace, answer, gates, deps V
Lookup state
aliases
wording to keys
store turnover
D01 D02
vectors, one per entry
judges ×3,
trained on the catalog
Check outcomes H
key (id(c), V(dep(c)))
date_dim @0
store_sales → date_dim @0
item @0
item @1
6
item, 'Y' @1
6
3
request ρ
Lookup
no SQL, no LLM call
Alias table
confirmed wording?
Recall
bge-m3, 8 nearest
Judge ×3
cross-encoder, p each
Decide
p
refer
.1
.9
hit
one definition
same name
all, with differences
refer
agent confirms; alias
miss
agent explores
4
hit: m, p, d
Check and run
one snapshot D_s
check C: outcome in H at V_s,
else run Q_c once
compile d(q0(D_s, p)) and run
SUM(ss_net_paid) ... WHERE
i_category = 'Electronics'
GROUP BY d_moy ORDER BY ...
features: per (u_t, t_i)
t_i
output
answer to the agent
6
outcomes at V, rev 1
Maintain
after each
write
Write
stamp V(T) per table
item: batch 0 → 1
insert-only: rows ΔT
Stage 1: versions
skip c if V(dep(c)) is unchanged
rerun 2 of 5
each once, for all definitions
Stage 2: inserted rows
check only ΔT, by index
key 17 now twice
Repair or retire
sole single edit, answer kept
i_is_current='Y' → rev 1
else retire; new value: pending
ETL job
loads item history
5
item
5 batch 0 → 1
i_item_sk
i_category
i_is_current
17
'Electronics'
'Y'
current
17
'Electronics'
'N'
new
store_sales
batch 0
date_dim
batch 0
Database
versions V(T)
change on writes
snapshot D_s